

Lab Exercise — Module 1

Introducing SIEM: Wazuh Architecture & Log Sources

Duration: 60 minutes | Instructor: Abhishek | Day 1

The Scenario:

You have just joined a company as a junior SOC analyst. Your first task: get a new server into the SIEM so the team can see its logs. The server is running Ubuntu 22.04 and has no monitoring agent installed yet. No one will know if something bad happens on this server until *you* connect it to Wazuh.

What You Will Accomplish

By the end of this lab, you will have:

- Explored all three Wazuh components on the live dashboard
- Installed and registered a Wazuh agent on an Ubuntu VM
- Confirmed log flow from the agent to the Manager
- Generated a real security event and found it as an alert
- Written your first WQL queries to search and filter alerts
- Added a custom log source path to the agent configuration

Prerequisites

- Wazuh Manager already running (provided by instructor)
- One Ubuntu 22.04 VM assigned to you (your target endpoint)
- Network connectivity between your VM and the Wazuh Manager
- Browser access to the Wazuh Dashboard URL

Lab Environment

Wazuh Manager IP:	<provided by instructor>
Dashboard URL:	https://<manager-ip>:443
Dashboard Username:	admin
Dashboard Password:	<provided by instructor>
Your VM IP:	<your assigned IP>
Agent config (Linux):	/var/ossec/etc/ossec.conf
Agent logs:	/var/ossec/logs/ossec.log

Task 1 — Explore the Three Wazuh Components (10 min)

Before connecting anything, let us understand what is already running.

Step 1.1 — Log into the Dashboard

1. Open your browser and go to the Dashboard URL above.
2. Log in with the credentials provided by the instructor.
3. You will land on the **Overview** page.

Step 1.2 — Find the Manager

1. In the left sidebar: click **Server Management** → **Overview**.
2. Note down: Manager hostname and Wazuh version.
3. The Manager config file on the server is at:
`/var/ossec/etc/ossec.conf`

Step 1.3 — Find the Indexer

1. Click **Indexer Management** in the left sidebar.
2. Check that cluster health shows **green**.
3. Note: alerts are stored in index pattern `wazuh-alerts-4.x-*`

Checkpoint: You should see Manager = Connected, Indexer = Green, Dashboard = Active.

Question: In your own words, what is the difference between the Manager and the Indexer?

Your answer: _____

Task 2 — Install the Wazuh Agent on Your Ubuntu VM (15 min)

This is the key task. Without an agent, your server is invisible to the SIEM.

Step 2.1 — Add the Wazuh repository and install the agent

Run these commands on your Ubuntu VM as root or with sudo:

```
# Step 1: Import the Wazuh GPG signing key
curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH \
  | gpg --no-default-keyring \
      --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg \
      --import

# Step 2: Add the Wazuh apt repository
echo "deb [signed-by=/usr/share/keyrings/wazuh.gpg] \
  https://packages.wazuh.com/4.x/apt/ stable main" \
  | sudo tee /etc/apt/sources.list.d/wazuh.list

# Step 3: Update package lists and install
sudo apt-get update
sudo apt-get install wazuh-agent -y
```

What just happened? You installed a small program on this server. Right now it does not know where the Wazuh Manager is. The next step fixes that.

Step 2.2 — Point the agent at your Manager

The agent config file is at: `/var/ossec/etc/ossec.conf`

```
# Open the config file
sudo nano /var/ossec/etc/ossec.conf

# Find this block near the top:
# <server>
#   <address>MANAGER_IP</address>
# </server>

# Replace MANAGER_IP with the actual Manager IP address provided
```

Save the file: **Ctrl+O**, then **Enter**, then **Ctrl+X**.

Step 2.3 — Start the agent and check its status

```
sudo systemctl daemon-reload
sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent
sudo systemctl status wazuh-agent
```

Checkpoint: The status output must show active (running).
If it shows failed — the most common reason is a wrong Manager IP in `ossec.conf`.
To check agent connection logs: `tail -f /var/ossec/logs/ossec.log`

Task 3 — Confirm the Agent Appears in the Dashboard (10 min)

Step 3.1 — Find your agent

1. In the Dashboard: click **Agents** from the left menu.
2. Search for your VM hostname or IP address.
3. The status column should show **Active** (green dot).

If the agent shows Pending or Never connected:

Run `sudo /var/ossec/bin/agent-auth -m <manager-ip>` on your VM to force registration,
then restart: `sudo systemctl restart wazuh-agent`

Step 3.2 — What log files is this agent already monitoring?

On your VM, check the default monitored paths:

```
# See all <localfile> blocks - these are the paths being monitored
grep -A3 "<localfile>" /var/ossec/etc/ossec.conf
```

Write down the log paths you see:

- Path 1: _____
- Path 2: _____
- Path 3: _____

Task 4 — Generate a Real Security Event (10 min)

Now let us create something suspicious on purpose and watch it appear as an alert.

Step 4.1 — Trigger a failed SSH login

Run this command from your VM. It will fail intentionally — that is the point:

```
# Attempt SSH with a non-existent username (this will fail)
ssh thisuserdoesnotexist@localhost
# When asked for a password, press Enter or type anything
```

Step 4.2 — Find the alert in the Dashboard

1. Go to **Security Events** in the Dashboard.
2. Filter by your agent name (click the agent name in the Agents list).
3. Look for an alert with **Rule ID 5710** — “SSH: Attempt to login using a non-existent user”.
4. Click the alert to expand it.

Checkpoint: You should see the failed SSH attempt as an alert, with the full raw log line visible at the bottom of the alert detail panel.

Record from the alert:

Rule ID: _____ Rule Level: _____ Agent Name: _____

Raw log line (copy it here):

Step 4.3 — Try a brute-force simulation (5 failed logins)

```
# Run 5 failed SSH attempts in a row
for i in {1..5}; do ssh wronguser@localhost; done
```

Now go back to the Dashboard. You should see a **new, higher-level alert** fire — Wazuh detects the pattern of repeated failures as a brute-force attempt.

Record: What is the Rule ID and level of the brute-force alert?

Rule ID: _____ Level: _____ Description: _____

Task 5 — Practice WQL Queries (10 min)

A SOC analyst who cannot search quickly is a slow SOC analyst. WQL is your superpower.

Open the **Security Events** page. Use the search bar at the top to run each query:

Step 5.1 — Basic filters

```
# Find all alerts at level 5 or above from your agent
rule.level>=5 AND agent.name:<your-agent-name>

# Find all authentication failure events
rule.groups:authentication_fail

# Find all SSH-related alerts
rule.groups:sshd
```

Step 5.2 — Field-specific searches

```
# Find events that mention a specific IP (replace with your VM IP)
data.srcip:192.168.x.x

# Find a specific Rule ID
rule.id:5710

# Find all alerts in the last 15 minutes
# (use the time picker in the top-right corner, not WQL)
```

Challenge: Write a single WQL query that finds all authentication failures at level 7 or above from your specific agent.

Your query: _____

How many results did you get? _____

Task 6 — Add a Custom Log Source (10 min)

Your server runs Nginx web server. Right now Wazuh is NOT monitoring its logs. Fix that.

Step 6.1 — Check if Nginx is installed

```
# Check if Nginx is running
sudo systemctl status nginx
```

If not installed:

```
sudo apt-get install nginx -y
sudo systemctl start nginx
```

Step 6.2 — Verify the Nginx log path exists

Nginx writes logs to these paths by default:

- Access log: /var/log/nginx/access.log
- Error log: /var/log/nginx/error.log

```
# Verify the log files exist
ls -lh /var/log/nginx/
```

Step 6.3 — Add the log paths to the Wazuh agent config

```
sudo nano /var/ossec/etc/ossec.conf
```

Add these two blocks **inside the** `<ossec_config>` **section**:

```
<!-- Nginx Access Log -->
<localfile>
  <log_format>apache</log_format>
  <location>/var/log/nginx/access.log</location>
</localfile>

<!-- Nginx Error Log -->
<localfile>
  <log_format>apache</log_format>
  <location>/var/log/nginx/error.log</location>
</localfile>
```

Step 6.4 — Restart the agent and generate a web event

```
# Restart agent to load the new config
sudo systemctl restart wazuh-agent

# Generate an Nginx log event (access the web server locally)
curl http://localhost/

# Generate a 404 error (file not found)
curl http://localhost/this-page-does-not-exist
```

Step 6.5 — Find the Nginx alert in the Dashboard

Go to **Security Events**, filter by your agent, and search:

```
rule.groups:web
```

Checkpoint: You should see HTTP request events from Nginx appearing as alerts. The 404 request should trigger a “File not found” rule.

Record: What Rule ID fired for the 404 request?

Rule ID: _____ Description: _____

Reflection Questions

Answer in your own words (3–4 sentences each). Think carefully — these questions will be discussed in class.

Q1. You have 200 Linux servers. Each one writes about 5,000 log lines per minute. That is 1,000,000 lines per minute across the fleet. Without a SIEM, how would you even begin to detect an attack? What would you miss?

Q2. A new web server was added to the network but no one configured the Wazuh agent. An attacker exploits it three weeks later. The SOC team has no alerts. Whose fault is this? What process would prevent this from happening again?

Q3. Your company has 10 network switches that cannot run a Wazuh agent. How would you get their logs into Wazuh? Write the exact config change you would make (hint: think about Syslog and the <remote> block in ossec.conf).

Quick Reference: Key Log Source Paths

Log Path / Channel	OS	What it Contains
/var/log/auth.log	Linux	SSH, sudo, logins
/var/log/syslog	Linux	General system events
/var/log/kern.log	Linux	Kernel messages
/var/log/nginx/access.log	Linux	HTTP requests
/var/log/nginx/error.log	Linux	Web server errors
/var/ossec/logs/ossec.log	Linux	Wazuh agent own log
Security	Windows	Logins, privilege events
System	Windows	Service/driver events
Microsoft-Windows-Sysmon/Operational	Windows	Process, network events
Microsoft-Windows-PowerShell/Operational	Windows	PowerShell execution

Submission Checklist

Before you leave, confirm you have:

- ☐ Screenshot of your agent showing **Active** status in the Dashboard
- ☐ Screenshot of the failed SSH alert (Rule ID 5710)
- ☐ Screenshot of the brute-force alert (higher level than 5710)
- ☐ Screenshot of an Nginx log event in the Dashboard
- ☐ Your WQL challenge query and the number of results
- ☐ Written answers to the three reflection questions